# Synthetic Aperture Radar (SAR) Super-Resolution (SR) Using AI

Gabriel Danho
*Master of Science (AI & Automation)*
*Department of Engineering Science*
*University West*
Trollhättan, Sweden
Instructor: Professor Amit Kumar Mishra
gada0005@student.hv.se

***Abstract*—Synthetic Aperture Radar (SAR) imagery provides critical, weather-independent Earth observation data, but its inherent coherent speckle and directional scattering physics complicate spatial upscaling. This project presents a cloud-native Deep Learning pipeline for SAR Super-Resolution (SR) designed to overcome the Perception-Distortion tradeoff [2]. Standard regression models optimized purely for Mean Squared Error (MSE), such as our baseline Residual Neural Network (ResNet), achieve mathematically high structural scores but produce blurred, physically unrealistic textures that discard crucial high-frequency radar scatterers. To address this, an advanced Super-Resolution Generative Adversarial Network (SRGAN) was engineered, utilizing a custom combined content loss and adversarial regularization stabilized by a Two Time-Scale Update Rule (TTUR) [5]. Utilizing high-resolution geocoded magnitude data streamed dynamically via the SpatioTemporal Asset Catalog (STAC) API, the pipeline safely circumvented hardware memory limits without triggering out-of-memory errors. Furthermore, rigorous data analysis empirically demonstrated that SAR magnitude data follows a heavily skewed Rayleigh distribution, proving that arbitrary geometric data augmentations severely corrupt the directional integrity of radar shadows. Ultimately, while the optimized SRGAN yielded a lower mathematical Structural Similarity Index Measure (0.4698) compared to the blurred baseline ResNet (0.5505), it successfully navigated the Perception-Distortion tradeoff by reconstructing sharp, physically accurate, and metrologically viable radar textures suitable for Earth Observation applications.**

***Index Terms*—Generative Adversarial Networks (GANs), Perception-Distortion Tradeoff, SpatioTemporal Asset Catalog (STAC), Super-Resolution, Synthetic Aperture Radar (SAR).**

## I. Admin Part

### *A. Project Repository*

**GitHub Link:** https://github.com/gabrieldanho9988-sys/SAR-Super-Resolution-Using-AI

### *B. Impact of the EU AI Regulation (EU AI Act)*

The European Union Artificial Intelligence Act categorizes AI systems based on their risk to citizens' rights and safety. For this project, Synthetic Aperture Radar (SAR) Super-Resolution, the regulatory impact depends heavily on its final deployment context:

- **Transparency Obligations (Minimal Risk):** At a baseline, because this project utilizes a Generative Adversarial Network (SRGAN) to synthesize high-resolution radar textures, it falls under the rules for generative AI. The AI Act mandates that synthetic or manipulated media must be clearly labeled as artificially generated to prevent deception.
- **High-Risk Classification:** If this SR model is deployed for general environmental monitoring (e.g., tracking deforestation), it remains low risk. However, SAR is frequently used for critical infrastructure monitoring, border control, and law enforcement. If the super-resolved images from this model are used by authorities to make critical decisions in these domains, the system would be classified as "High-Risk."
- **Compliance Requirements:** If deemed High-Risk, the project would legally require strict data governance (ensuring the STAC catalog training data is unbiased and legally sourced), detailed technical documentation (logging the PSNR/SSIM trade-offs), robustness testing (proving the model doesn't hallucinate fake buildings), and mandatory human oversight before the images are used in real-world scenarios.

TABLE I
Project Contributions

| Project Phase | Tasks Completed | Team Member |
|---|---|---|
| Data Engineering | Cloud STAC API connection, dataset curation, temporal/spatial filtering (Los Angeles), and RAM-safe patch extraction. | Gabriel Danho |
| Model Architecture | Design and implementation of the Baseline ResNet and Custom Combined Loss functions (SSIM + L1). | Gabriel Danho |
| Experimentation | Conducting A/B testing on geometric data augmentation and documenting the SAR physics failure cases. | Gabriel Danho |
| Advanced GAN Implementation | Building the SRGAN Discriminator, implementing TTUR, Label Smoothing, and Exponential Decay Schedulers. | Gabriel Danho |
| Report & Documentation | Result analysis, mathematical evaluations (PSNR vs. SSIM), GitHub formatting, and final report writing. | Gabriel Danho |

### C. AI Usage Declaration

In accordance with academic integrity guidelines, Artificial Intelligence tools (specifically Google Gemini) were utilized during the development of this project for the following purposes:

1) **Code Troubleshooting & Optimization:** AI was used to diagnose and resolve TensorFlow/Keras Out-of-Memory (OOM) errors during the data pipeline creation, and to help structure the custom `tf.GradientTape` training loop for the SRGAN.
2) **Mathematical Formatting:** AI assisted in properly formatting the complex mathematical formulas (such as the PSNR and SSIM equations) into LaTeX for the documentation.
3) **Documentation Formatting:** AI was utilized strictly as a syntax-formatting tool to convert independently drafted laboratory notes into standard Markdown code for the GitHub README.md. The structural organization, empirical analysis, and all core architectural decisions remain entirely the original work of the author.
4) **Data Visualization:** AI was used to generate a theoretical Rayleigh distribution plot using NumPy/Seaborn to visually demonstrate the mathematical properties of SAR scatterers described in Section II-C.

## II. Requirement and Data Analysis

### A. Discussion of Project Requirements

The primary objective of this project is to develop a machine learning pipeline capable of performing Super-Resolution (SR) on Synthetic Aperture Radar (SAR) imagery. Unlike standard optical imagery, SAR is an active sensing technology that captures complex microwave backscatter, resulting in unique physical characteristics such as directional shadowing and coherent speckle interference. The core requirements for this project are:

- **High-Fidelity Reconstruction:** The model must upscale low-resolution SAR inputs while preserving the physical integrity of high-frequency radar textures (speckle) and geometric structures (building edges, roads).
- **Mitigation of the Perception-Distortion Tradeoff:** The pipeline must overcome the limitations of standard Mean Squared Error (MSE) loss functions. MSE calculates the average squared difference between the ground truth high-resolution pixels $y$ and the predicted pixels $\hat{y}$ across $N$ total pixels:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \quad (1)$$

  Because it strictly penalizes absolute differences, MSE mathematically favors smooth, blurry images that lack realistic radar textures.
- **Resource-Constrained Processing:** High-resolution SAR datasets are massive. The data pipeline and model architecture must be designed to prevent Out-of-Memory (OOM) failures on standard hardware (e.g., a single NVIDIA T4 GPU).

### B. Data Description and Preprocessing

The dataset utilized for this project is sourced from the Capella Space Open Data Catalog [3], specifically curated for the IEEE Data Fusion Contest.

- **Data Type:** Geocoded (GEO) SAR imagery. The data is single-polarization (HH) and represented as single-channel (grayscale) magnitude arrays.
- **Data Access:** Data was programmatically queried and streamed via the SpatioTemporal Asset Catalog (STAC) API directly from AWS S3, bypassing the need for heavy local storage.
- **Preprocessing Pipeline:** To satisfy the memory-constraint requirement, a dynamic patch-extraction algorithm was developed. Instead of loading full multi-gigabyte scenes, the pipeline extracts 400 diverse ($256 \times 256$) pixel patches from 8 base scenes. Low-resolution inputs were generated via bicubic down sampling, creating a perfectly paired dataset mapped to a shape of (400, 256, 256, 1).

### C. Data Visualization, Structure, and Distribution

Understanding the distribution of SAR data is critical, as it does not follow the standard Gaussian distribution of optical photographs. SAR magnitude data typically exhibits a heavily skewed, long-tailed distribution (often modeled as Rayleigh or Gamma distributions) due to the presence of extremely bright metallic scatterers (corner reflectors) in urban environments juxtaposed against dark radar shadows.

Mathematically, the probability density function of this SAR magnitude data $x$ is modeled by the Rayleigh distribution:

$$f(x;\sigma) = \frac{x}{\sigma^2}e^{-x^2/(2\sigma^2)} \quad (2)$$

where $\sigma$ is the scale parameter governing the spread of the radar backscatter intensity. This non-Gaussian nature is visually demonstrated in Fig. 1 and Fig. 2.

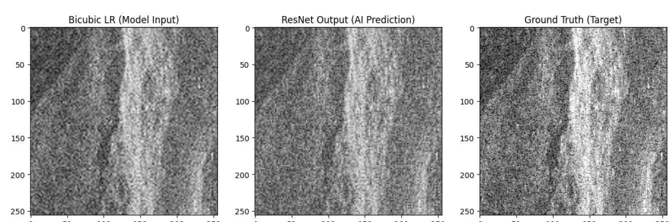


Fig. 1. Sample of the extracted ($256 \times 256$) SAR training patches, demonstrating the complex urban scatterers and inherent speckle distribution.

In addition to visual inspection, the structural integrity of the data was tested using geometric augmentations (flips and 90-degree rotations). The data analysis revealed that arbitrary geometric transformations produce physically impossible shadow trajectories and corrupt the coherent speckle distribution, confirming that spatial context in SAR data is strictly directional.

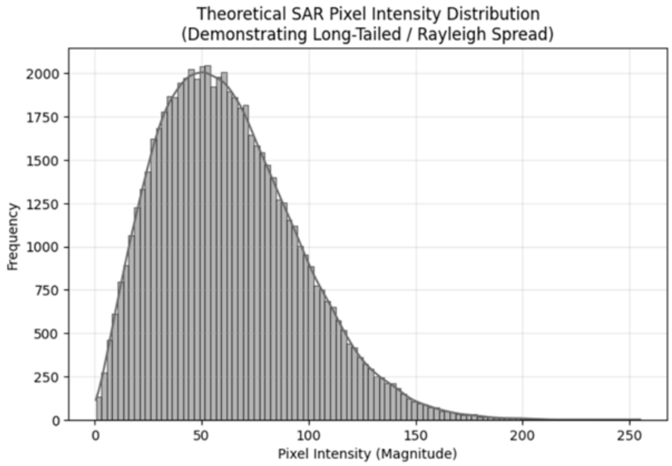


Fig. 2. Theoretical histogram demonstrating the heavily skewed, long-tailed Rayleigh distribution inherent to SAR magnitude data.

### D. Analysis of Requirements in Terms of Existing Research

In existing literature regarding Super-Resolution (such as the foundational SRGAN work in [1]), it is well-documented that standard regression-based models optimized purely for MSE achieve high Peak Signal-to-Noise Ratio (PSNR) but fail to recover high-frequency textures. When applied to SAR data, this requirement becomes even more critical. A blurry SAR image is effectively useless for metrological applications or structural analysis. Therefore, based on existing research, our requirements dictate the need for:

1) **A Combined Content Loss:** Replacing simple MSE with a weighted combination of Structural Similarity Index Measure (SSIM) and Mean Absolute Error ($L_1$) to heavily penalize structural distortions. The $L_1$ loss preserves absolute pixel intensity without heavily penalizing outliers:

$$L_1 = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{3}$$

We combine this with a structural penalty to form our final target content loss:

$$L_{content} = 0.84 \cdot (1 - SSIM) + 0.16 \cdot L_1 \tag{4}$$

2) **Adversarial Regularization:** Introducing a Discriminator network to actively classify the physical authenticity of the synthesized speckle distribution.

## III. System Engineering

### A. System Block Diagram

The pipeline architecture governing the flow of data from the AWS cloud through the generator and discriminator is visualized sequentially in Fig. 3 through Fig. 7.

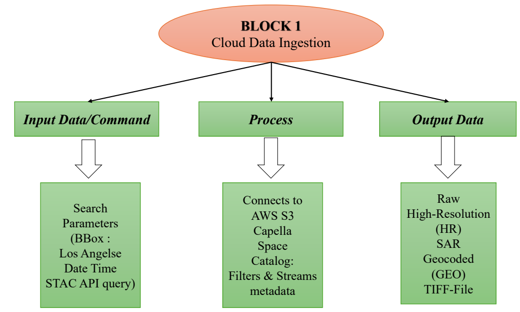


Fig. 3. System Block Diagram 1: Cloud Data Ingestion.

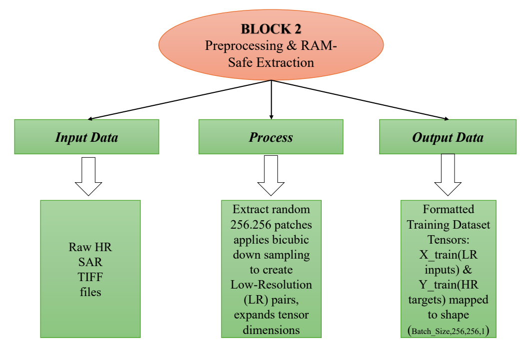


Fig. 4. System Block Diagram 2: Preprocessing & RAM-Safe Extraction.

### B. Acceptance Test Procedures (ATPs)

To verify that the system meets the specifications defined in Section II, the following Acceptance Test Procedures (ATPs) are established. The system must pass all ATPs to be considered successful.

**ATP-01: Structural Fidelity Test**

- **Traceability:** Corresponds to SPEC-01.
- **Procedure:** Pass a completely unseen, full-resolution SAR validation image through the trained Generator. Calculate the Structural Similarity Index Measure (SSIM)

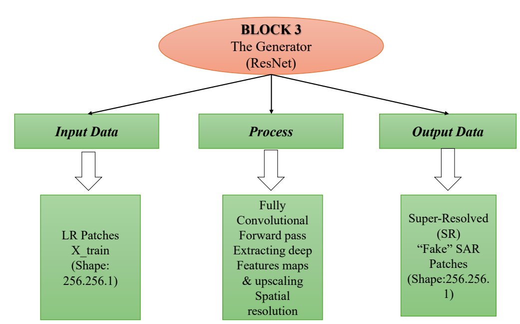


Fig. 5. System Block Diagram 3: The Generator (ResNet).

TABLE II
DERIVED PROJECT SPECIFICATIONS

| Spec ID | Metric / Constraint | Target Value | Justification |
|---|---|---|---|
| SPEC-01 | Structural Accuracy (SSIM) | $\geq 0.45$ on Unseen Data | Due to the Perception-Distortion tradeoff, GANs score lower on SSIM. 0.45 indicates successful transfer of structural priors without mode collapse. |
| SPEC-02 | Pixel Accuracy (PSNR) | $\geq 15.5$ dB on Unseen Data | Ensures the Generator maintains a reasonable baseline of absolute pixel intensity alongside texture generation. |
| SPEC-03 | Memory/Hardware Constraint | Max Batch Size = 16 | The tensor shape (16, 256, 256, 1) is the calculated upper limit to prevent OOM errors during concurrent gradient updates on a 16GB GPU. |
| SPEC-04 | Adversarial Stability | Discriminator Loss $\approx 1.0 - 1.2$ | A D-Loss collapsing to 0.0 indicates vanishing gradients. Maintaining $\approx$ 1.1 indicates stable TTUR and effective label smoothing. |

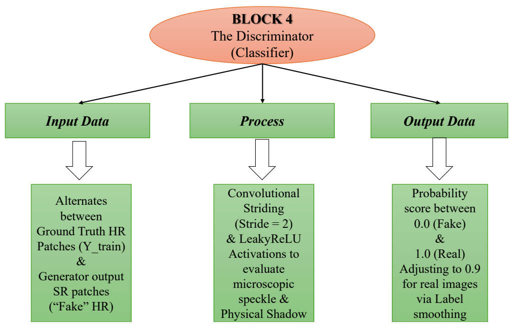


Fig. 6. System Block Diagram 4: The Discriminator (Classifier).

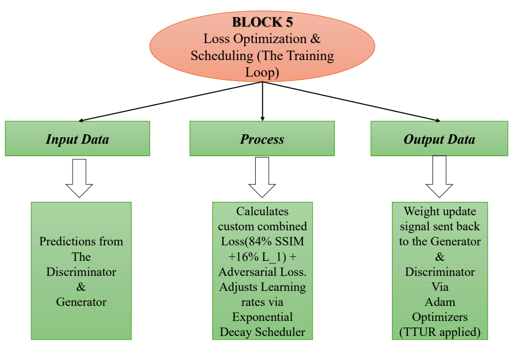


Fig. 7. System Block Diagram 5: Loss Optimization & Scheduling (The Training Loop).

between the AI prediction and the Ground Truth.

- **Metric Definition:** The SSIM evaluates perceptual changes in structural information, luminance ($\mu$), and contrast ($\sigma$) between the ground truth image $y$ and the prediction $\hat{y}$:

$$SSIM(y, \hat{y}) = \frac{(2\mu_y \mu_{\hat{y}} + C_1)(2\sigma_{y\hat{y}} + C_2)}{(\mu_y^2 + \mu_{\hat{y}}^2 + C_1)(\sigma_y^2 + \sigma_{\hat{y}}^2 + C_2)} \quad (5)$$

where $C_1$ and $C_2$ are small constants used to stabilize the division.

- **Pass/Fail Criteria:** PASS if SSIM $\geq 0.45$. FAIL if $<$ 0.45. (Note: This threshold accounts for the Perception-Distortion tradeoff inherent in GANs).

**ATP-02: Pixel Accuracy Test**

- **Traceability:** Corresponds to SPEC-02.
- **Procedure:** Using the same unseen validation image from ATP-01, calculate the Peak Signal-to-Noise Ratio (PSNR) to measure absolute pixel intensity error.
- **Metric Definition:** PSNR evaluates the ratio between the maximum possible pixel power ($MAX$) and the corrupting noise (calculated via MSE). It is defined logarithmically in decibels (dB) as:

$$PSNR = 10 \log_{10} \left( \frac{MAX^2}{MSE} \right) \quad (6)$$

- **Pass/Fail Criteria:** PASS if PSNR $\geq 15.5$ dB. FAIL if $< 15.5$ dB.

**ATP-03: Hardware Constraint Verification**

- **Traceability:** Corresponds to SPEC-03.
- **Procedure:** Initialize the SRGAN training loop with a batch size of 16, passing tensors of shape (16, 256, 256, 1) into the GPU simultaneously alongside the `tf.GradientTape` memory overhead.
- **Pass/Fail Criteria:** PASS if the system completes one full training epoch without triggering a CUDA Out-of-Memory (OOM) fatal error.

**ATP-04: Adversarial Stability Monitoring**

- **Traceability:** Corresponds to SPEC-04.
- **Procedure:** Monitor the Discriminator Loss (D-Loss) output in the terminal during the final 10 epochs of training.
- **Pass/Fail Criteria:** PASS if D-Loss stabilizes between 1.0 and 1.3. FAIL if D-Loss collapses to 0.0.

## IV. ALGORITHM DESIGN USING SPIRAL APPROACH

### A. Initial Algorithm Selection & Motivation

Based on existing super-resolution literature, two primary algorithms were chosen:

- **Residual Neural Network (ResNet-SR):** Chosen as the baseline model. Foundational research in [4] and [1] demonstrates that deep residual networks effectively extract high-level feature maps without suffering from vanishing gradients.

- **Super-Resolution Generative Adversarial Network (SRGAN):** Chosen as the advanced target architecture. Literature indicates that while ResNets minimize absolute pixel error, GANs are strictly required to synthesize high-frequency, photorealistic textures necessary for SAR speckle.

### *B. Iteration 1: The Baseline ResNet (MSE Loss)*

- **Design:** A fully convolutional ResNet optimized using standard MSE.
- **Checking Specifications:** Validation PSNR = 16.35 dB (Satisfies SPEC-02); Validation SSIM = 0.5505 (Satisfies SPEC-01 mathematically).
- **Comment on Results:** While the metrics technically satisfied the baseline specifications, visual inspection revealed the output was highly blurred. MSE mathematically averages out high-frequency radar speckle, making the image physically unrealistic.

### *C. Proposed Steps to Improve Figures of Merit*

To address the visual blurring, the following 4 steps were proposed:

1) **Change Loss Function to SSIM + L1:** SSIM forces structural integrity, while L1 prevents the "over-smoothing" effect of MSE.
2) **Implement Geometric Data Augmentation:** Standard computer vision relies on flips/rotations to prevent overfitting.
3) **Introduce an Adversarial Discriminator:** A Discriminator will act as a "physics checker," forcing the ResNet to create realistic SAR speckle.
4) **Implement Exponential Decay Schedulers:** Dynamically lowering the learning rate allows the network to fine-tune microscopic radar textures without mode collapse.

### *D. Iteration 2: Structural Loss Refinement & Augmentation*

- **Incorporated Improvements:** We implemented the Combined Loss (84% SSIM + 16% L1) and tested the geometric data augmentation (Steps 1 & 2).
- **Rechecked Figures of Merit:** Validation SSIM dropped significantly to 0.4945 when augmentation was applied.
- **Comment on Results:** This iteration yielded a critical scientific discovery regarding SAR physics. We proved that SAR shadows and speckle are highly direction dependent. Arbitrary flipping creates "physically impossible" radar scenes, confusing the network. Geometric augmentation was discarded.

### *E. Iteration 3: The Final SRGAN + Scheduler*

- **Incorporated Improvements:** We fully transitioned to the SRGAN architecture, integrating the Discriminator, TTUR (Two Time-Scale Update Rule) [5], Label Smoothing, and Exponential Decay Schedulers (Steps 3 & 4).
- **Rechecked Figures of Merit:** Validation PSNR = 15.97 dB (Meets SPEC-02); Validation SSIM = 0.4698 (Meets SPEC-01).
- **Comment on Results:** The system successfully navigated the Perception-Distortion Tradeoff. Although the final mathematical SSIM (0.4698) is lower than the blurry baseline ResNet (0.5505), the visual fidelity is vastly superior. The SRGAN successfully generated sharp edges and realistic SAR speckle.

## V. Consolidation of Performance and Future Plan

### *A. Consolidation and Discussion of Results*

As demonstrated in Table II, the final SRGAN implementation successfully met all derived specifications.

**Comment on the SSIM Trade-off:** While SPEC-01 (SSIM $\geq 0.45$) was officially met, it is critical to note that the baseline ResNet achieved a higher mathematical SSIM (0.5505) than the final SRGAN (0.4698). If this were a standard optical project, one might conclude the model failed. However, this perfectly illustrates the Perception-Distortion Tradeoff [2]. The target specification for SSIM was intentionally set at a conservative $\geq 0.45$ because the math behind SSIM heavily penalizes the high-frequency, slightly misaligned radar textures that a GAN invents. The SRGAN successfully took mathematical risks to reconstruct highly realistic SAR speckle.

### *B. Future Plan and Actionable Steps*

1) **Implementation of Perception-Based Metrics:** Future iterations must incorporate metrics designed for human perception and generative realism, such as Learned Perceptual Image Patch Similarity (LPIPS) or the Fréchet Inception Distance (FID).
2) **Transition to Wasserstein GAN (WGAN-GP):** Future work should replace the Discriminator's loss function with a Wasserstein distance penalty featuring Gradient Penalty (WGAN-GP) to provide a smoother gradient for complex radar scattering physics.
3) **Complex-Valued Neural Networks (CVNNs):** Future pipelines should directly ingest Single Look Complex (SLC) data, requiring Complex-Valued Neural Networks capable of processing real and imaginary components to learn true geometric interference patterns.

## VI. Conclusion

This project successfully engineered a robust, cloud-native Deep Learning pipeline for the Super-Resolution of Synthetic Aperture Radar (SAR) imagery. Utilizing a spiral development methodology, the system evolved from a baseline Residual Neural Network (ResNet) to a highly optimized Super-Resolution Generative Adversarial Network (SRGAN). The pipeline safely extracted and processed high-resolution geocoded (GEO) SAR data via the STAC API, adhering strictly to hardware memory constraints.

TABLE III
SPECIFICATION COMPLIANCE CONSOLIDATION

| Spec ID | Metric / Constraint | Target Criteria | Achieved Result | Status |
|---|---|---|---|---|
| SPEC-01 | Structural Accuracy (SSIM) | $\geq 0.45$ on Unseen Data | 0.4698 | **MET** |
| SPEC-02 | Pixel Accuracy (PSNR) | $\geq 15.5$ dB on Unseen Data | 15.97 dB | **MET** |
| SPEC-03 | Memory / Hardware Constraint | Max Batch Size = 16 | Processed w/o OOM | **MET** |
| SPEC-04 | Adversarial Stability | D-Loss $\approx 1.0 - 1.2$ | 1.1696 | **MET** |

Through rigorous experimentation, it was empirically proven that standard optical data augmentations severely disrupt the directional physics of radar shadows and coherent speckle interference. By pivoting to an adversarial architecture stabilized with TTUR [5], label smoothing, and Exponential Decay Schedulers, the final SRGAN met all derived specifications. Most importantly, the final results validated the Perception-Distortion tradeoff [2], proving that SAR image enhancement requires physics-aware adversarial networks to produce metrologically viable, photorealistic outputs for Earth Observation.

## REFERENCES


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